

RA 12009 Compliant

Enterprise Procurement Management System

Complete Installation & Deployment Manual

Version 1.0 — Generated July 03, 2026
Enterprise Procurement Management System

Table of Contents

1. System Overview
2. Hardware & Software Requirements
3. Local Development Installation
4. Database Setup & Sample Data
5. Running the Application
6. Docker Installation
7. Production Deployment (AWS)
8. Test Accounts
9. Verification & Testing
10. Troubleshooting
11. Additional Documentation

1. System Overview

The Enterprise Procurement Management System (PMS) is a full lifecycle platform for Philippine Government Agencies, compliant with **RA 12009** (New Government Procurement Act). It covers procurement from the DBM General Appropriations Act (GAA) through payment, with complete audit trail, configurable workflows, and role-based access control.

Procurement lifecycle:

GAA → Budget Allocation → APP → PPMP → Purchase Request → CAF → BAC Procurement → PhilGEPS Posting → Bidder Portal → Bidding → Award → Notice to Proceed → Purchase Order → Delivery → Inspection → Acceptance → Payment → Reports & Analytics

Technology stack: Laravel 12, PHP 8.3, Livewire 3, TailwindCSS 4, AlpineJS, MySQL 8 (SQLite for local sandbox), AWS S3, Laravel Queue & Scheduler, Spatie Permission, DomPDF, Laravel Excel, UUID primary keys, Service Layer + Event-Driven Architecture.

2. Hardware & Software Requirements

Minimum (local development)

- PHP 8.3+ with extensions: pdo_mysql (or pdo_sqlite), mbstring, bcmath, intl, gd, zip, fileinfo
- Composer 2.x
- Node.js 20+ and npm
- MySQL 8.0+ or SQLite (zero-config sandbox)
- 4 GB RAM recommended; 2 GB minimum

Recommended (production)

- MySQL 8.0 Multi-AZ (RDS) or equivalent managed database
- Redis for cache/queue at scale
- AWS S3 private bucket for document storage
- SMTP relay for email notifications (NOA, NTP, CAF, PhilGEPS alerts)
- HTTPS termination (ALB + ACM certificate)

3. Local Development Installation

Step 1 — Clone and install dependencies

```
git clone <repo-url> pms-procurement
cd pms-procurement
composer install
npm install
```

Step 2 — Environment configuration

```
cp .env.example .env
php artisan key:generate
```

Edit .env and configure at minimum:

Variable	Local sandbox	MySQL production-like
DB_CONNECTION	sqlite	mysql
DB_DATABASE	database/database.sqlite	pms_procurement
SESSION_DRIVER	file	database or redis
CACHE_STORE	file	database or redis
DOCUMENTS_DISK_DRIVER	local	s3
APP_URL	http://localhost:8000	https://your-agency.gov.ph

For SQLite: touch database/database.sqlite. Match APP_URL to your dev-server port (e.g. :8001).

Step 3 — Build frontend assets

```
php artisan storage:link
npm run build
```

For active development with hot reload: npm run dev

4. Database Setup & Sample Data

Option A — Migrations & seeders (recommended)

```
php artisan migrate --seed
```

This creates all tables and seeds:

- 19 roles / 102 permissions
- Organizational structure (departments, divisions, offices, cost centers)
- Reference data (modes of procurement, fund sources, UACS, PAPs, holidays)
- 18 internal test accounts + 1 Bidder Portal account
- Sample FY2026 procurement walkthrough (GAA → CAF)

Option B — Restore from SQL dump (SQLite)

A pre-exported database snapshot is included at database/pms_procurement.sql.

```
rm -f database/database.sqlite
sqlite3 database/database.sqlite < database/pms_procurement.sql
```

Note: this SQL dump is SQLite-specific and is not directly importable into MySQL.

5. Running the Application

All-in-one dev stack (recommended)

```
composer run dev
```

Starts: web server (:8000), queue worker, log tiler (Pail), and Vite dev server.

Individual processes

```
php artisan serve
php artisan queue:listen
php artisan schedule:work
```

Access URLs

Portal	URL
Internal agency login	http://localhost:8000/login
Bidder / Supplier portal	http://localhost:8000/bidder/login
Health check	http://localhost:8000/up

6. Docker Installation

The repository includes a production-shaped Docker Compose stack.

Service	Role
app	PHP-FPM 8.3 — migrations on boot, serves application
nginx	HTTP termination, proxies to app:9000
queue	php artisan queue:work (notifications, exports)
scheduler	Runs schedule:run every 60 seconds
mysql	MySQL 8.0 primary datastore
redis	Optional cache/queue backend
mailhog	Local SMTP catcher (UI at :8025)

```
cp .env.example .env
# Set DB_HOST=mysql, MAIL_HOST=mailhog, MAIL_PORT=1025
docker compose build
docker compose up -d
docker compose exec app php artisan db:seed
```

Visit **http://localhost:8080** (nginx). MailHog: **http://localhost:8025**.

7. Production Deployment (AWS)

Recommended topology (see docs/AWS_DEPLOYMENT.md for full details):

- **Route 53** — DNS for agency domain
- **ALB + ACM** — HTTPS load balancing
- **ECS Fargate** — app, queue, and scheduler services (same Docker image)
- **RDS MySQL 8.0** — Multi-AZ database
- **S3** — private, versioned, encrypted document bucket
- **ElastiCache Redis** — optional cache/queue at scale

Production .env highlights:

- APP_ENV=production, APP_DEBUG=false
- DB_CONNECTION=mysql pointing to RDS
- DOCUMENTS_DISK_DRIVER=s3 with AWS credentials
- QUEUE_CONNECTION=database or redis
- DEMO_ACCOUNTS_VISIBLE=false

8. Test Accounts

Default password for all seeded accounts: **Passw0rd!2026** (local/UAT only).

Role	Email
Super Admin	superadmin@pms.gov.ph

System Admin	sysadmin@pms.gov.ph
BAC Chairperson	bac.chair@pms.gov.ph
BAC Secretariat	bac.secretariat@pms.gov.ph
Budget Officer	budget.officer@pms.gov.ph
Planning Officer	planning.officer@pms.gov.ph
HOPE	hope@pms.gov.ph
Division Chief	division.chief@pms.gov.ph
End User	end.user@pms.gov.ph
Cashier	cashier@pms.gov.ph
Internal Auditor	auditor@pms.gov.ph
Viewer	viewer@pms.gov.ph
Bidder / Supplier	bidder@supplier.com (portal)

Demo accounts appear on login pages when DEMO_ACCOUNTS_VISIBLE=true (default outside production).

9. Verification & Testing

```
composer test
# or: php artisan test
```

28+ feature tests cover RBAC, budget integrity (no overallocation), and the full procurement lifecycle (GAA → CAF and BAC → Payment).

Post-install smoke check:

1. Sign in as Super Admin
2. Open Dashboard, GAA, APP, PPMP, Purchase Requests
3. Verify sample FY2026 data exists after seeding
4. Sign in to Bidder Portal and confirm opportunities list loads

10. Troubleshooting

Symptom	Solution
Vite manifest not found	Run <code>npm run build</code> or keep <code>npm run dev</code> running
MySQL connection refused	Start MySQL or switch to SQLite for local dev
Login hangs / too many attempts	Use <code>SESSION_DRIVER=file</code> and <code>CACHE_STORE=file</code> with SQLite; run <code>php artisan cache:clear</code>
Redirect to wrong port after login	Set <code>APP_URL</code> to match dev-server port (e.g. <code>http://localhost:8001</code>)
Alpine multiple instances warning	Do not call <code>Alpine.start()</code> in <code>app.js</code> — Livewire handles Alpine
403 on all pages	Re-run <code>php artisan db:seed --class=PermissionSeeder</code>
Emails not sending	Set <code>MAIL_MAILER=log</code> or configure SMTP / MailHog

11. Additional Documentation

Document	Location
Architecture & ERD	docs/ARCHITECTURE.md
Docker Guide	docs/DOCKER.md
AWS Deployment	docs/AWS_DEPLOYMENT.md
Security	docs/SECURITY.md
RBAC Roles & Permissions	docs/RBAC.md
Backup & Restore	docs/BACKUP_RESTORE.md
API Reference	docs/API.md
In-app Help & User Manuals	Help module (after login)